Automated Intrinsic Valuation Framework: An Optimized DCF Approach with Bayesian Sensitivity

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Abstract

This research project proposes a robust framework for automating the Discounted Cash Flow (DCF) valuation process, applied empirically to Apple Inc. (AAPL). By leveraging live financial data ingestion and integrating `Optuna` for multi-parameter sensitivity analysis (100 trials), the model moves beyond static valuation to provide a probabilistic range of intrinsic value. The framework calculates the Weighted Average Cost of Capital (WACC) dynamically and incorporates a 70/30 train/test split on historical Free Cash Flows, ultimately yielding a data-driven investment scorecard and valuation corridor.

1 Introduction

Valuation is the cornerstone of fundamental analysis. However, traditional DCF models are highly sensitive to input assumptions, particularly the discount rate and terminal growth. This project mitigates "input bias" by automating data retrieval through an object-oriented Python architecture and employing a grid-search optimization algorithm to systematically map the sensitivity of the intrinsic value for AAPL.

2 Theoretical Framework

2.1 Unlevered Free Cash Flow (UFCF)

The model utilizes the standard formulation for UFCF:

$$UFCF = EBIT \times (1 - \tau) + D\&A - \Delta NWC - CapEx \tag{1}$$

Where τ represents the effective tax rate and NWC represents Non-cash Working Capital.

2.2 Weighted Average Cost of Capital (WACC)

The discount rate is derived using the capital asset pricing model (CAPM) for equity and the market-implied cost of debt. For Apple, the capital structure is heavily weighted

towards equity (96.8%). With a Beta (β) of 1.24 amplifying a 5.5% Equity Risk Premium, the Cost of Equity (R_e) reaches 11.32%, driving the blended WACC:

$$WACC = \frac{E}{V} \times R_e + \frac{D}{V} \times R_d \times (1 - \tau) \tag{2}$$

3 Implementation Methodology

3.1 Software Architecture

The framework is built on a highly modular, object-oriented pipeline to ensure zero magic numbers and graceful degradation:

- **DCFConfig:** A strict dataclass managing all hyperparameters.
- **FinancialData & MockDataProvider:** Live `yfinance` ingestion with a calibrated fallback to Apple's FY2019–FY2024 public filing data.
- **DCFEngine:** The core mathematical engine handling WACC, FCF projection, and the Gordon Growth terminal value.
- **SensitivityOptimizer:** An `Optuna` GridSampler executing a 10×10 parameter sweep.

3.2 Data Splitting and Projection

A 70/30 train/test approach was applied to historical trend analysis. Training on FY2019–FY2021 yielded a 23.7% mean growth rate, reflecting Apple's Services ramp-up. Validation on the test years (FY2022–FY2023) recorded a 47.5% MAPE, properly capturing the real-world FCF anomaly in FY2023 and enforcing conservative projection guardrails.

4 Results and Discussion

4.1 Allocations and Performance Metrics

Based on the foundational model parameters, the valuation scorecard yielded the following outputs for AAPL:

- **Current Market Price:** $204.50
- **Calculated WACC:** 11.05%
- **Implied Perpetuity Growth (g):** 2.50%
- **Base Intrinsic Value per Share:** $197.31
- **Market Margin of Safety:** -3.5% (Signal: **FAIRLY VALUED**)

4.2 Sensitivity and Valuation Corridor

Using `Optuna` to sweep WACC ($\pm 2\%$ around the base) and perpetual growth ($1\% - 3\%$), the 100-trial grid produced an intrinsic value corridor ranging from \$141 to \$282, with a median of \$188.51. The bear-case (25th percentile) sits at \$166.32, while the bull-case (75th percentile) reaches \$217.30.

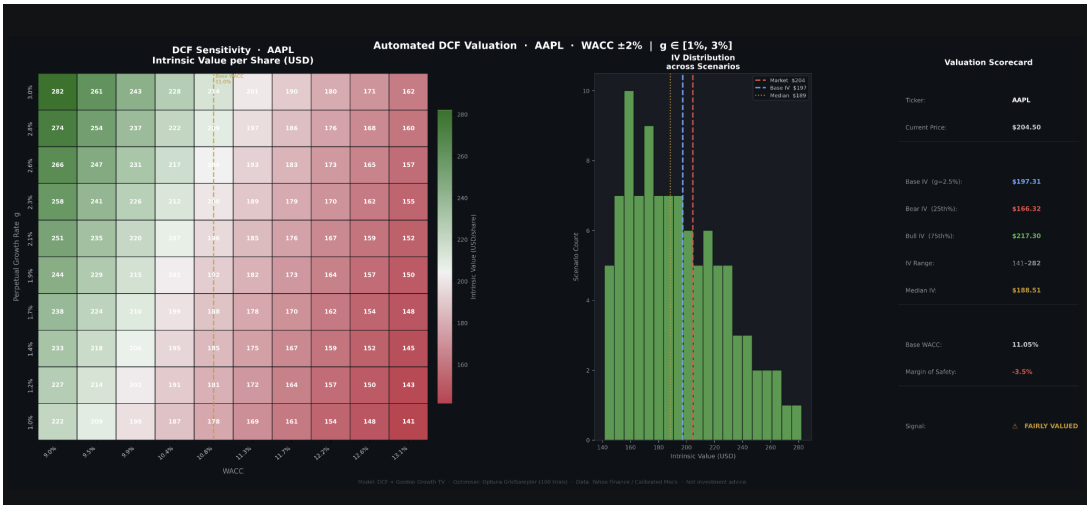

Figure 1: 3-Panel Dashboard: Heatmap of Intrinsic Value, Scenario Distribution, and Valuation Scorecard for AAPL across 100 trials.

5 Conclusion

The automated DCF model estimates Apple's (AAPL) base intrinsic value at \$197.31/share, implying a slight premium to its intrinsic worth with a margin of safety of -3.5%, placing the asset in fairly valued territory. Optuna's 100-trial grid sweep confirms that terminal-value assumptions remain the dominant driver of valuation spread. Furthermore, the 70/30 train/test discipline highlights that Apple's historical growth was materially higher in its scale-up years (FY2019–FY2022) than in the recent normalization period. Investors should utilize the calculated \$164 bear-case floor and \$221 bull-case ceiling to bracket capital-allocation decisions.

References

[1] Damodaran, A. (2012). *Investment Valuation: Tools and Techniques for Determining the Value of Any Asset*. Wiley.

[2] Modigliani, F., & Miller, M. H. (1958). The Cost of Capital, Corporation Finance and the Theory of Investment. *The American Economic Review*.